**DE LA SALLE UNIVERSITY – DASMARIÑAS**
**COLLEGE OF INFORMATION AND COMPUTER STUDIES**
**INFORMATION TECHNOLOGY DEPARTMENT**

# COURSE SYLLABUS
**Second Semester, SY 2024-2025**

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|---|---|---|
| **COURSE CODE** | **:** | **S-ITPC224** |
| **COURSE TITLE** | **:** | **Networking 1** |
| **COURSE TYPE** | **:** | **Lecture** |
| **COURSE CREDIT** | **:** | **Two Units** |
| **LEARNING MODALITY** | **:** | **Traditional Blended Learning Modality** |
| **PRE/ CO-REQUISITES** | **:** | **None** |
| **CO/REQUISITES** | **:** | **S-ITPC224LA** |
| **PROFESSOR** | **:** | **Michael Angelo F. Manalo** |
| **CONSULTATION HOUR** | **:** | **Friday 11:00am-12:00nn ; COS 100C** |
| **CONTACT INFORMATION** | **:** | **mfmanalo@dlsud.edu.ph ; 09664056138** |

## VISION
A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

## MISSION
To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

## COURSE DESCRIPTION:

Deals with computer network principles covering different layers, topologies and models of computer networks (explore Open System Interconnection), network programming, and network applications. This course also focuses on the TCP/IP protocol, network architecture (including implementation, administration and maintenance) and infrastructure design. Students also gain "hands-on" networking experience by installing and setting up a network operating system, building a small LAN, and expected to do systems/network programming and make use of simulation and measurement tools to gain an appreciation of current network design and performance issues.

## COURSE LEARNING OUTCOMES:

By the end of the semester, the students can:

CLO1. explain the concept of network communication;
CLO2. create a simulated network using Packet Tracer;
CLO3. build a functioning network;
CLO4. explain the principles of IPv4 and IPv6 address management;

CLO5. configure initial settings on a network device;
CLO6. configure basic network security;
CLO7. describe data communications and network models, topologies, protocols, standards and architectures; and
CLO8. explain how to use security best practices to mitigate attacks.

**LEARNING PLAN:**

**Learning Outcomes for Midterm Period**

| Course Learning Outcome | Topic Learning Outcome |
|---|---|
| CLO1. Explain the concept of network communication. | TLO1. Explain the concept of network communication. |
| CLO2. Create a simulated network using Packet Tracer. | TLO2. Explain the requirements for getting online. |
| CLO3. Build a functioning network. | TLO3. Explain the importance of standards and protocols in network communications. |
| CLO4. Explain the principles of IPv4 and IPv6 address management. | TLO4. Explain the features of an IP address. |

**Midterm Period:**

| Module No and Learning Outcome(s) | Week<br><br>Date/s | Teaching-Learning Lessons and Activities / Assessment Strategies | Technology Enabler | Onsite | Asynchronous | Allotted Hours |
|---|---|---|---|---|---|---|
| Module No. 1<br>CLO 1<br>TLO 1 | Week 1<br>Jan. 20-25 | Gospel Reading, Reflection and Prayer<br>Presentation of LMS Expectations:<br>Netacad Course Experience Interface, Submission, Around the Online Module<br>Schoolbook Details for Submission of Assessment<br>Module 1 Communications in a Connected World<br>Download Packet Tracer version 8.0.1 | • Schoolbook<br>• Microsoft 365<br>• MSTeams<br>• Netacad | √ | | 2.0 |

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|---|---|---|---|---|---|---|
| Module No. 1<br>CLO 2<br>TLO 1 | Week 2<br>Jan. 27-<br>Feb. 1 | Gospel Reading, Reflection and Prayer<br>Module 1 Discussion: Communicating on a Local Network, Key Terms and Concepts<br>Interactive Assessment: Match Protocol Terms to Definitions, and Determine the MAC Address of a Host, My Protocol Rules, IP Addresses and Network Communications.<br>Show How to use Packet Tracer for the first time<br>Enabling Assessment 1 – Create a simple network | • Schoolbook<br>• MSTeams<br>• Netacad | | √ | 2.0 |
| Module No. 2<br>CLO 3,4<br>TLO 2,3,4 | Week<br>3<br>Feb 3-8 | Gospel Reading, Reflection and Prayer<br>Module 2 – Online Connections<br>Enabling Assessment 2 – My On-Line Day/My Local Network Share the screen cap of your achievement for the day | • Schoolbook<br>• MSTeams<br>• Netacad | √ | | 2.0 |
| Module No. 3<br>CLO 1<br>TLO 1 | Week 4<br>Feb 10-<br>15 | Gospel Reading, Reflection and Prayer<br>Read Module 3: Explore Networks with Packet Tracer<br>Enabling Assessment 3 – Use Packet Tracer using Physical View | • Schoolbook<br>• MSTeams<br>• Netacad | | √ | 2.0 |
| Module No. 4<br>CLO 1<br>TLO 2 | Week 5<br>Feb 17-<br>22 | Gospel Reading, Reflection and Prayer<br>Module 4 Discussion: Network Addressing for IPv4<br>Try the Online Binary Game<br>Enabling Assessment 4: Network Address Computation | • Schoolbook<br>• MSTeams<br>• Netacad | √ | | 2.0 |
| Module No. 4<br>CLO 1,2<br>TLO 1,2 | Week 6<br>Feb 24 -<br>Mar. 1 | Gospel Reading, Reflection and Prayer<br>Continue Reading remaining part of Module 4:<br>Fixed Length Subnetting for IPv4<br>Network Addressing:<br>and<br>Enabling Assessment 5: IP Address Task | • Schoolbook<br>• MSTeams<br>• Netacad | | √ | 2.0 |

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|---|---|---|---|---|---|---|
| Module No. 4<br>CLO 3<br>TLO 3 | Week 7<br>Mar. 3 -<br>Mar. 8 | Gospel Reading, Reflection and Prayer<br>Continue Reading remaining part of Module 4: Network Addressing:<br>Variable Length Subnet Mask for IPv4 | • Schoolbook<br>• MSTeams<br>• Netacad | √ | | 2.0 |
| Module No. 4<br>CLO 4<br>TLO 4 | Week 8<br>Mar. 10-<br>14 | Gospel Reading, Reflection and Prayer<br>Continue Reading remaining part of Module 4: Network Addressing:<br>Variable Length Subnet Mask for IPv4 | • Schoolbook<br>• MSTeams<br>• Netacad | | √ | 2.0 |
| | Week 9<br>Mar. 15-<br>21 | Summative Assessment: via Netacad. | • Schoolbook<br>• MSTeams<br>• Netacad | √ | | 2.0 |
| Total | | | | 10.0 | 8.0 | 18.0 |

**Learning Outcomes for Final Period**

| **Course Learning Outcome** | **Topic Learning Outcome** |
|---|---|
| CLO5. Configure initial settings on a network device. | TLO5. Explain how to use the Cisco IOS. |
| CLO6. Configure basic network security. | TLO6. Compare different types of network connections. |
| CLO7. Describe data communications and network models, topologies, protocols, standards and architectures. | TLO7.Explain how software tools can mitigate network security threats. |
| CLO8. Explain how to use security best practices to mitigate attacks. | TLO8. Explain how software tools can mitigate network security threats |

| **Module No and Learning Outcome(s)** | **Week** | **Teaching-Learning Lessons and Activities / Assessment Strategies** | **Technology Enabler** | **Onsite** | **Asynchronous** | **Allotted Hours** |
|---|---|---|---|---|---|---|
| Module No. 5<br>CLO<br>TLO | Week 10<br>Mar. 24-<br>29 | Gospel Reading, Reflection and Prayer<br>Read Module 5: Communication Principles<br>Choosing a Transport Protocol<br>Transport Layer Terminology | • Schoolbook<br>• MS Teams | √ | | 2.0 |

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|---|---|---|---|---|---|---|
| Module No. 6<br>CLO<br>TLO | Week 11<br>Mar. 31-<br>Apr. 5 | Gospel Reading, Reflection and Prayer<br>Discussion of Module 6: Network Design and Access Layer<br>ISP Connection Terms<br>Enabling Assessment 1 PT Configure Basic Wireless Security | • Schoolbook<br>• MS Teams | | √ | 2.0 |
| Module No. 7<br>CLO 5,6<br>TLO 6 | Week 12<br>April.7-<br>12 | Gospel Reading, Reflection and Prayer<br>Discussion of Module 7: Network Security<br>Security Threats, Virus, Worm, or Trojan Horse, Computer Threats and Identify the Purpose of Each Security Tool<br>Enabling Assessment 2 Interactive Assessment on Network Security | • Schoolbook<br>• MS Teams<br>• Netacad | √ | | 2.0 |
| Module No.<br>CLO<br>TLO | Week 13<br>April.<br>14-19 | Independent Learning Week<br>Check the independent learning hub posted for this purpose.<br>Holy Week Break | • Schoolbook<br>• MS Teams<br>• Netacad | | √ | 2.0 |
| Module No. 7<br>CLO 6<br>TLO 6 | Week 14<br>April.<br>21 – 26 | Gospel Reading, Reflection and Prayer<br>Discussion of Module 7: Network Security<br>Security Threats, Virus, Worm, or Trojan Horse, Computer Threats and Identify the Purpose of Each Security Tool<br>Enabling Assessment 3 – Configure a Firewall | • Schoolbook<br>• MS Teams<br>• Netacad | √ | | 2.0 |
| Module No. 7<br>CLO 7<br>TLO 7 | Week 15<br>April. 28-<br>May 2 | Gospel Reading, Reflection and Prayer<br>Continue Reading remaining part of Module 7<br>Try the interactive exercises and do online netacad chapter quiz and exam<br>Try also to Configure Firewall Settings | • Schoolbook<br>• MS Teams<br>• Netacad | | √ | 2.0 |

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|---|---|---|---|---|---|---|
| Module No. 8<br>CLO 7<br>TLO 7 | Week 16<br>May. 4-10 | Gospel Reading, Reflection and Prayer<br>Discussion of Module 8: Configuring Cisco Devices Enabling Assessment<br>Connect to a Switch via the Console Port<br>Navigating the IOS<br>Enabling Assessment 4 Using the Cisco IOS Show Commands<br>Implementing Basic Connectivity | • Schoolbook<br>• MS Teams<br>• Netacad | √ | | 2.0 |
| Module No. 8<br>CLO 8<br>TLO 8 | Week 17<br>May. 12-17 | Gospel Reading, Reflection and Prayer<br>Read Module 8: Testing and Troubleshooting<br>Discover the systematic troubleshooting process and approach:<br>Using Network Utilities<br>Physical Connectivity<br>Enabling Assessment 5 A Wireless Connection | • Schoolbook<br>• MS Teams<br>• Netacad | | √ | 2.0 |
| | Week 18<br>May. 19-24 | Summative Assessment: via NetAcad | • Schoolbook<br>• MS Teams<br>• Netacad | √ | | 2.0 |
| Total | | | | 8.0 | 10.0 | 18.0 |

**GRADING SYSTEM:**

| Midterm | % | Final | % |
|---|---|---|---|
| Enabling Assessments (Activities, Quizzes, Presentations, etc.) | 50% | Enabling Assessments (Activities, Quizzes, Presentations, etc.) | 50% |
| Summative Assessment (Midterm Examination) | 30% | Summative Assessment (Final Paper) | 30% |
| Class Participation | 20% | Class Participation | 20% |
| | 100% | | 100% |

**FINAL GRADE =** $\frac{\text{Midterm Grade + Final Grade}}{2}$

**COURSE POLICIES AND REQUIREMENTS:**

1. This Course follow the **Traditional Blended Learning Mode**. There will be 50% onsite and 50% asynchronous.

2. **Enrollment in an E-Class.** The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. **Communication and Feedback.**

    a. **Onsite**. Classes shall happens as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.

    b. **Online**. Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communications platform.

        i. **Forum** – at the Forum Tab at Schoolbook, student can post their general concern. The forum is public, and any post can be seen by the class.

        ii. **Messaging** – at Schoolbook. Any private or confidential matters may be messaged through the schoolbook messaging feature.

        iii. **Email** – you mail email your professor (Professor's email will be given during the course orientation)

    c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat messages. No selling

4. **Attendance Policy**.

    a. **Attendance Policy for Onsite Classes.** Is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. **Intellectual Property Policy.**

    a. **Policy and copyright infringement** will not be tolerated. Any violation of such will earn a student a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submission to avoid copyright infringement and/or plagiarism.
    d. Giving of passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWFO).

6. **Other Policies.**

    a. **Attire –** students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. **Students must be honest at all times**; cheating and plagiarism will merit a grade of 0.00.

c. **Cellular/Mobile phones should always be in silent mode during class hours**; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot be used as calculator during examination.
d. Any complaints against the teacher or against classmate should be properly addressed to the subject teacher for appropriate action. Students may seek help and guidance of their academic/ registration adviser in resolving the issue with the subject teacher when such mediation is needed.

**REFERENCES:**


| Call number or e-provider | Reference Material |
|---|---|
| ISBN-10: 0-13-816610-2 | Academy, C. N. C. N. (2023). *It Essentials Companion Guide V8*. Cisco Press. |
| ISBN 9780367420161 | Grant, A. E., & Meadows, J. H. (2020). *Communication Technology update and Fundamentals*. Routledge. |
| ISBN:9781119650744 | Panek,C. (2020). Networking Fundamentals. John Wiley & Sons. |
| QA 76.73.P98 .Or82 2019 | Sarker, Dr. M. 2019. Learning Python networking : a complete guide to build and deploy strong networking capabilities using Python 3.7 and Ansible. Second edition. UK : Packt Publishing, Ltd. |


**ONSITE REFERENCES**

**ONLINE REFERENCES**

| Online Module | Reference Material |
|---|---|
| www.netacad.com | Cisco Networking Academy, https://www.netacad.com/portal/resources/course-resources/ccna-srwe (Week 4-6, 11, 13-16) |
| skillsforall.com | Networking Basics https://skillsforall.com/course/networking-basics?userLang=en-US (Week 4-6, 11, 13-16) |

**VIDEOS**

| Link | Title |
|---|---|
| https://contenthub.netacad.com/devnet/1.3.1 | The Power of Code (Week 1-2) |
| https://contenthub.netacad.com/srwe-dl/12.4.1 | CAPWAP Operation (Week 17) |
| https://contenthub.netacad.com/srwe-dl/12.6.1 | WLAN Threats (Week 17) |
| https://contenthub.netacad.com/srwe-dl/12.7.1 | Secure WLANs (Week 17) |

Prepared:

**Information Technology Department**
Second Semester, SY 2024-2025

Endorsed:

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